# Constraints on Higgs-charm couplings at CMS

Chirayu Gupta
on behalf of the CMS Collaboration

IIHE, Vrije Universiteit Brussel (VUB)

Higgs 2026, Louvain-la-Neuve, 28 October 2026

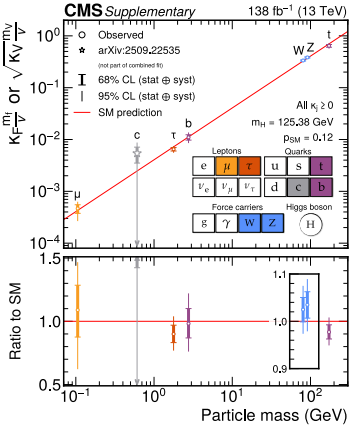
# Why the Higgs-charm coupling?

- Higgs couplings to the 3rd generation (t, b, $\tau$) and to $\mu$ agree with the SM
- **2nd-generation quark Yukawas are the open frontier**: charm is the most accessible
  - SM: $\mathcal{B}(\mathrm{H} \to \mathrm{c\bar{c}}) \approx 2.9\%$, about 20× below $\mathrm{H} \to \mathrm{b\bar{b}}$
  - Huge QCD backgrounds, c-jets hard to tell apart from b- and light jets
- BSM: flavour-compatible models allow $|\kappa_\mathrm{c}|$ of order 3

  e.g. spontaneous flavour violation, vector-like quarks

  (arXiv:2410.05236)

  $\Rightarrow$ current sensitivity is now entering this region
- Is the Higgs mechanism also responsible for the 2nd-generation quark masses?




Rep. Prog. Phys. 89 (2026) 087801 (supplementary); $\kappa_\mathrm{c}$ from PRL 136 (2026) 011801

$$\kappa_\mathrm{c} = y_\mathrm{c} / y_\mathrm{c}^\mathrm{SM} \quad (\text{SM: } \kappa_\mathrm{c} = 1)$$

# How CMS probes $\kappa_c$

## Direct $H \to c\bar{c}$ (rate $\propto \kappa_c^2/\Gamma_H$)

| | |
|---|---|
| VH($H \to c\bar{c}$) | PRL 131 (2023) 061801 |
| $t\bar{t}H(H \to c\bar{c})$ + VH comb. | PRL 136 (2026) 011801 |
| Boosted $H \to c\bar{c}$ (mostly ggH) | PRL 131 (2023) 041801 |

best: $|\kappa_c| < 3.5$ (2.7 exp.)

## H + charm production (cg $\to$ cH)

| | |
|---|---|
| cH($H \to \gamma\gamma$) | JHEP 11 (2025) 060 |
| cH($H \to WW$) + comb. | JHEP 08 (2026) 051 |

best: $|\kappa_c| < 47$ (51 exp.)

## Indirect: H production and width

| | |
|---|---|
| $\gamma$H + on-shell $H \to 4\ell$ | PRD 112 (2025) 112001 |
| Off-shell + on-shell $H \to 4\ell$ **NEW** | arXiv:2607.23352 |

$\kappa_c \in [-3.7, 3.2]$ to $[-8.6, 7.5]$ (by assumptions)

## Rare exclusive decays

| | |
|---|---|
| $H \to J/\psi\gamma$, $\psi(2S)\gamma$ | PLB 865 (2025) 139462 |

sensitive to the *sign* of $\kappa_c$; details in J. Bierkens' talk

$-166 < \kappa_c < 208$

All results with 13 TeV Run 2 data, up to 138 $\mathrm{fb}^{-1}$. **NEW** = public in 2026.

# Direct $\mathrm{H} \to \mathrm{c\bar{c}}$: VH and boosted H

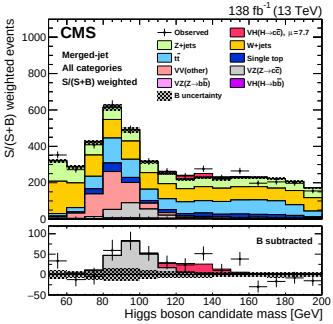
- **VH**: leptonic W/Z, resolved + merged-jet (ParticleNet) topologies
- VZ(Z→c$\bar{\text{c}}$) observed, 5.7$\sigma$

**VH($\mathrm{H} \to \mathrm{c\bar{c}}$), 138 fb$^{-1}$**

$\mu < 14\,(7.6)$
$1.1 < |\kappa_\mathrm{c}| < 5.5$ $\quad(|\kappa_\mathrm{c}| < 3.4$ exp.)

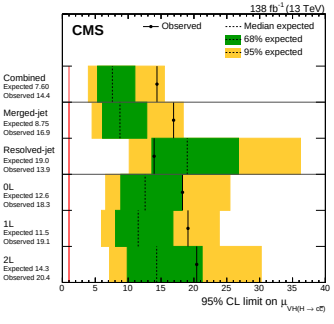
**Boosted $\mathrm{H} \to \mathrm{c\bar{c}}$, $p_\mathrm{T} > 450$ GeV**

$\mu < 47\,(39)$ (mostly ggH)




VH merged-jet: $m(\mathrm{H_{cand}})$, S/(S+B) weighted

PRL 131 (2023) 061801; PRL 131 (2023) 041801

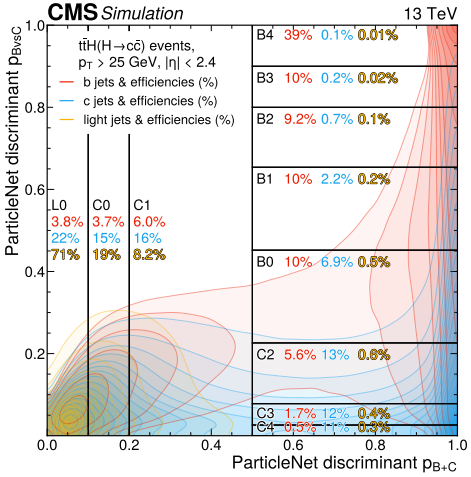



VH: $\mu$ limits per channel

# $\mathrm{t\bar{t}H(H \to c\bar{c})}$: a new channel

PRL 136 (2026) 011801, CMS-HIG-24-018

- First search for $\mathrm{H \to c\bar{c}}$ in $\mathrm{t\bar{t}H}$; 0, 1, 2 lepton $\mathrm{t\bar{t}}$ decays
- Many jets of all flavours: ParticleNet b vs. c vs. light **2D categories**
- ParT event classifier separating $\mathrm{t\bar{t}H(cc)}$, $\mathrm{t\bar{t}H(bb)}$, $\mathrm{t\bar{t}Z}$, $\mathrm{t\bar{t}}$+HF
- **Simultaneous** fit of $\mathrm{t\bar{t}H(H \to c\bar{c})}$, $\mathrm{t\bar{t}H(H \to b\bar{b})}$, $\mathrm{t\bar{t}Z(Z \to b\bar{b}/c\bar{c})}$
  - $\mathrm{t\bar{t}}$+cc/bb normalised in dedicated regions
  - $\mathrm{t\bar{t}Z(Z \to c\bar{c})}$ as a standard candle: $\mu = 1.0 \pm 0.8$
- By-product: $\mathrm{t\bar{t}H(H \to b\bar{b})}$ $\mu = 0.91^{+0.26}_{-0.22}$, $4.4\sigma$ (4.5 exp.)



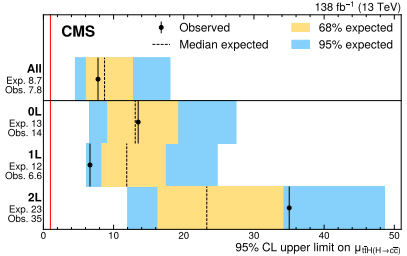

b, c, light jets in the ParticleNet (b vs. light, c vs. light) plane and the 2D tagging categories

# $\mathrm{t\bar{t}H(H \to c\bar{c})}$: results

## $\mathrm{t\bar{t}H(H \to c\bar{c})}$, 138 $\mathrm{fb}^{-1}$

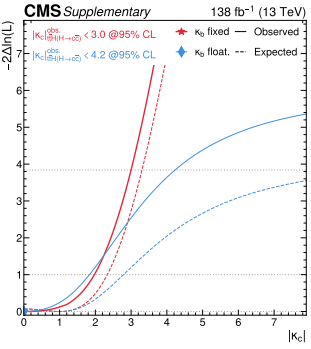
$\mu_{\mathrm{t\bar{t}H(cc)}} < 7.8\,(8.7)$
$\sigma\mathcal{B} < 0.11\,(0.13)$ pb
$|\kappa_\mathrm{c}| < 3.0\,(3.3)$, $\kappa_\mathrm{b} = 1$
$|\kappa_\mathrm{c}| < 4.2$ obs., $\kappa_\mathrm{b}$ floating

- Expected sensitivity comparable to $\mathrm{VH(H \to c\bar{c})}$
- Simultaneous bb/cc fit constrains $\kappa_\mathrm{b}$ and $\kappa_\mathrm{c}$ together




$\mu_{\mathrm{t\bar{t}H(cc)}}$ limits per channel

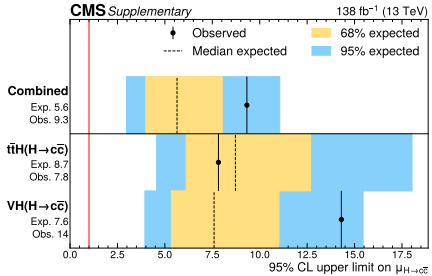
PRL 136 (2026) 011801, CMS-HIG-24-018

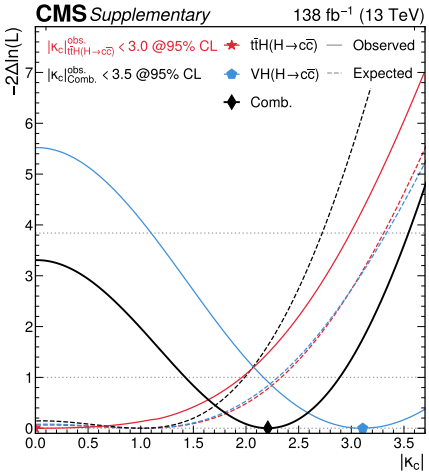



$|\kappa_\mathrm{c}|$ scan, $\kappa_\mathrm{b}$ fixed / floating

# Combination of ttH and VH: tightest constraint on $\kappa_c$

PRL 136 (2026) 011801, CMS-HIG-24-018




## ttH + VH

$\mu_{H\to c\bar{c}} < 9.3$ (5.6 exp.)

$|\kappa_c| < 3.5$ (2.7 exp.)

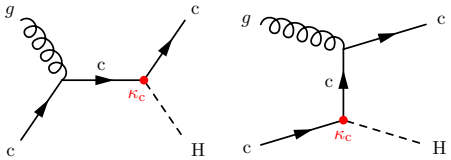
Expected limit improves by ∼20% over the best single channel

Most stringent direct constraint on $\kappa_c$ to date; expected reach now below the BSM-motivated $|\kappa_c| \sim 3$

# H + charm production: $\kappa_c$ in production

- $\kappa_c$ enters the **production** vertex: independent of $H \to c\bar{c}$-decay; clean H decays + a c-tagged jet
- $\sigma_{cH} = 90\,\text{fb}$: $\sigma\mathcal{B} \approx 0.2\,\text{fb}$ ($\gamma\gamma$), $\approx 20\,\text{fb}$ (WW)
- Challenges
  - H+c also arises from g→c$\bar{c}$ splitting in ggH: $\kappa_c$-**independent**, nearly identical final state
  - soft, often forward c-jets
  - flavour-scheme modelling of cH (→ T. Bevilacqua, next talk)
- $\kappa_c$ from "flat direction": other H rates kept SM-like

JHEP 11 (2025) 060; JHEP 08 (2026) 051




LO diagrams for cH production; $\kappa_c$ enters at the marked vertices

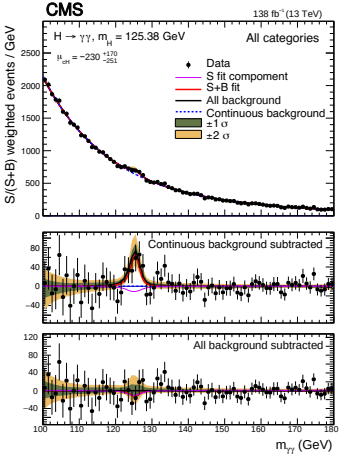
# cH(H$\to\gamma\gamma$)

JHEP 11 (2025) 060, CMS-HIG-23-010

- First search for cH with H$\to\gamma\gamma$, 138 $\mathrm{fb}^{-1}$
- Two BDTs, trained without c-tag inputs:
  - BDT1: cH vs. ggH (largest resonant background)
  - BDT2: cH vs. continuum $\gamma\gamma$+jets
- 9 categories per year (27 in total); signal from fits to $m_{\gamma\gamma}$

## Result, 95% CL

$\mu_{\mathrm{cH}} < 243\,(355)$
$|\kappa_{\mathrm{c}}| < 38.1\,(72.5)$



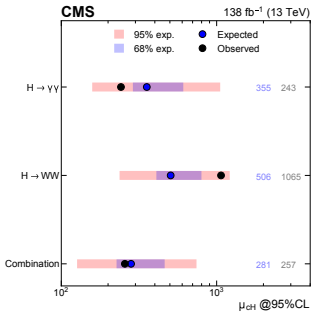

$m_{\gamma\gamma}$, all categories, S/(S+B) weighted

# cH(H→WW) and combination

- H→WW→e$\nu\mu\nu$ + c-jet; SRs with 1 or >1 c-jet, CRs for $t\bar{t}$ and high $m_{\ell\ell}$
- Two BDTs: vs. non-H backgrounds and vs. other H processes
- H+c background ($\kappa_c$-independent) split by jet flavour; correlation with cH: −87%
- 1-POI fit: $\mu_{\mathrm{cH}} < 1065\,(506)$, $|\kappa_c| < 211\,(95)$
  2-POI (H+c bkg floating): $< 1804\,(1097)$

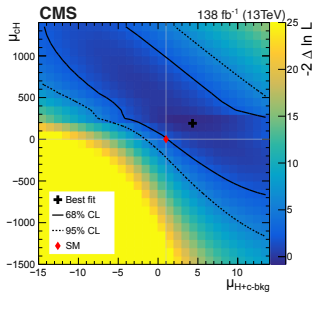
## $\gamma\gamma$ + WW combination

$\mu_{\mathrm{cH}} < 257\,(281)$
$|\kappa_c| < 47\,(51)$




$\mu_{\mathrm{cH}}$ limits: $\gamma\gamma$, WW, combined

JHEP 08 (2026) 051, CMS-HIG-24-009

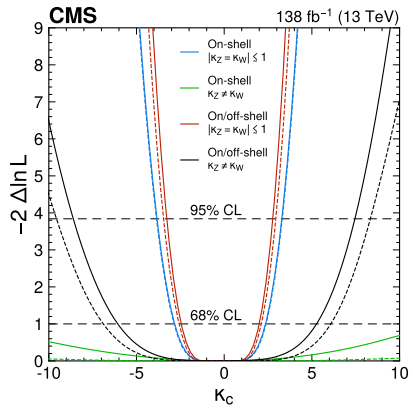



WW: $\mu_{\mathrm{cH}}$ vs. $\mu_{\mathrm{H+c\ bkg}}$

# Indirect constraints on $\kappa_c$

- Large light-quark Yukawas enhance $q\bar{q} \to H$ and $\gamma H$ production and change $\Gamma_H$
- **$\gamma$H + on-shell H**$\to 4\ell$: $\kappa_c \in [-3.7, 3.2]$ ($[-3.8, 3.3]$)
  other light Yukawas floating, $|\kappa_V| \leq 1$
- NEW **Off-shell + on-shell H**$\to 4\ell$: off-shell rate breaks the width degeneracy
  - no assumption on $\kappa_Z$, $\kappa_W$: $\kappa_c \in [-8.6, 7.5]$ ($[-9.6, 8.3]$)
  - with $|\kappa_V| \leq 1$: $\kappa_c \in [-3.3, 2.8]$ ($[-3.5, 2.9]$)
  - width measurement: see F. Errico's talk
- **H**$\to$**J/**$\psi\gamma$: $-166 < \kappa_c < 208$ (see J. Bierkens' talk)

PRD 112 (2025) 112001; arXiv:2607.23352; PLB 865 (2025) 139462




arXiv:2607.23352: $\kappa_c$ scans, on-shell and on+off-shell, with and without $|\kappa_V| \leq 1$

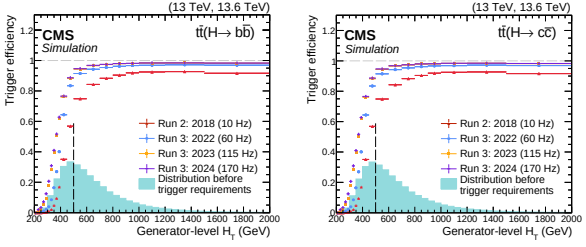
# Where we stand: CMS constraints on $\kappa_c$

| Probe | Reference | $\kappa_c$ at 95% CL: obs. (exp.) | Main assumptions |
|---|---|---|---|
| $t\bar{t}H + VH(H \to c\bar{c})$ | PRL 136 (2026) 011801 | $\lvert\kappa_c\rvert < 3.5$ (2.7) | $\kappa_b = 1$, SM-like width |
| $t\bar{t}H(H \to c\bar{c})$ | PRL 136 (2026) 011801 | $\lvert\kappa_c\rvert < 3.0$ (3.3) | $\kappa_b = 1$ |
| $VH(H \to c\bar{c})$ | PRL 131 (2023) 061801 | $1.1 < \lvert\kappa_c\rvert < 5.5$ (3.4) | $\kappa_b = 1$ |
| $\gamma H + H \to 4\ell$ | PRD 112 (2025) 112001 | $[-3.7, 3.2]$ ($[-3.8, 3.3]$) | $\lvert\kappa_V\rvert \leq 1$ |
| On+off-shell $H \to 4\ell$ | arXiv:2607.23352 | $[-8.6, 7.5]$ ($[-9.6, 8.3]$) | none on $\kappa_{Z,W}$ |
| cH ($\gamma\gamma$ + WW) | JHEP 08 (2026) 051 | $\lvert\kappa_c\rvert < 47$ (51) | |
| $H \to J/\psi\gamma$ | PLB 865 (2025) 139462 | $[-166, 208]$ | $\kappa_\gamma = 1$ |

- Direct $H \to c\bar{c}$ searches lead; indirect probes add sensitivity to the **sign** of $\kappa_c$ and different model assumptions
- Several probes now reach $|\kappa_c| \sim 3$, where flavour-compatible BSM scenarios live

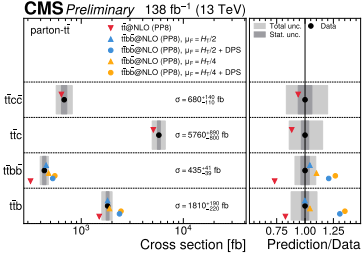
# Toward Run 3: tools already in place

## Heavy-flavour triggers




arXiv:2608.18331 (CMS-BTV-25-002): ML b/c tagging at the HLT; $\mathrm{tt}\mathrm{H}(\mathrm{H} \to \mathrm{c}\bar{\mathrm{c}})$ trigger efficiency, Run 2 vs. 2022–24

## $\mathrm{tt}+\mathrm{c}\bar{\mathrm{c}}$ background measured




CMS-PAS-TOP-26-003 **NEW**:

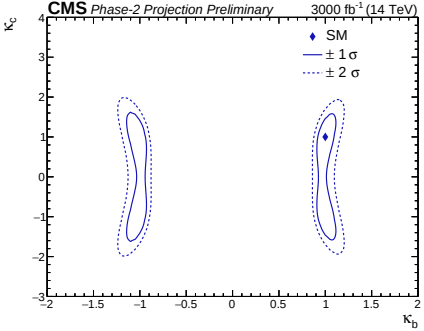
$$\sigma_{\mathrm{ttcc}} = 680^{+130}_{-100}\,(\mathrm{syst}) \pm 40\,(\mathrm{stat})\,\mathrm{fb}$$

- Run 3 at 13.6 TeV: dataset already larger than Run 2, new taggers (UParT, GloParT), parking and scouting streams for hadronic final states
- Run 3 analyses are under way in all channels

# Prospects at the HL-LHC

- ATLAS + CMS input to the European Strategy update
  (arXiv:2504.00672, CMS-HIG-25-002), $3\,\mathrm{ab}^{-1}$ per experiment:
  - VH($\mathrm{H} \to \mathrm{c\bar{c}}$): $\approx 1.6\sigma$
  - $|\kappa_\mathrm{c}| < 1.5$ at 95% CL, 30% better than previous projections
- History: expected $\mu_\mathrm{H} \to \mathrm{c\bar{c}}$ limit improved from $\sim$37 (2016 data) to 5.6 (Run 2 VH+$\mathrm{t\bar{t}H}$)
  far faster than luminosity alone: taggers, new channels, analysis techniques
- Further gains: Run 3 data, timing and track-trigger upgrades, ML tagging; theory modelling and tagger calibration become limiting

Is $\mathrm{H} \to \mathrm{c\bar{c}}$ evidence reachable at the HL-LHC?
Determining $|\kappa_\mathrm{c}|$ at $\mathcal{O}(1)$ is.




CMS-HIG-21-008 (supplementary): expected $(\kappa_\mathrm{b}, \kappa_\mathrm{c})$ contours at $3000\,\mathrm{fb}^{-1}$, 14 TeV, extrapolated from the Run 2 VH($\mathrm{H} \to \mathrm{c\bar{c}}$) analysis

# Summary

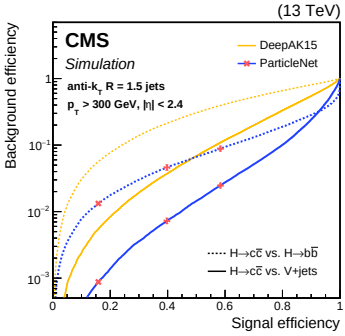
- The charm Yukawa is the most accessible 2nd-generation quark coupling, and CMS probes it in many complementary ways
- Direct searches: $\mathrm{t\bar{t}H(H \to c\bar{c})}$ combined with $\mathrm{VH(H \to c\bar{c})}$
  $\Rightarrow |\kappa_c| < 3.5$ **(2.7 exp.)**, the most stringent constraint to date
- H+c production probes $\kappa_c$ independently of $\mathrm{H \to c\bar{c}}$: $\gamma\gamma$ + WW give $|\kappa_c| < 47$ (51 exp.)
- Indirect probes ($\gamma$H, on/off-shell H$\to 4\ell$, H$\to$J/$\psi\gamma$) reach $|\kappa_c| \sim 3$ under different assumptions; off-shell H$\to 4\ell$ removes the assumption on $\kappa_{Z,W}$
- Sensitivity is now reaching the BSM-motivated region $|\kappa_c| \sim 3$
- Run 3: new HLT tagging and a measured $\mathrm{t\bar{t}+c\bar{c}}$ background; HL-LHC projects $|\kappa_c| < 1.5$

Thank you!

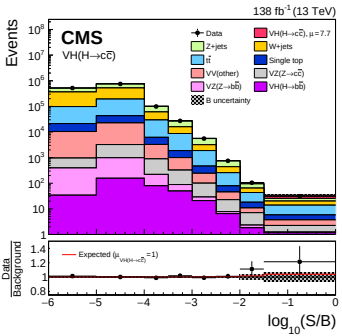
# Backup

# VH($\mathrm{H} \to \mathrm{c\bar{c}}$): event categories and sensitivity

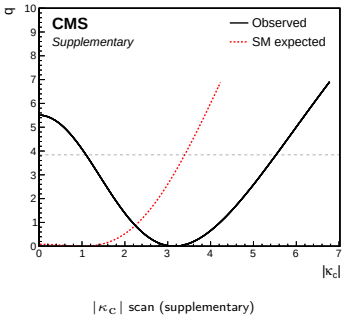
PRL 131 (2023) 061801




ParticleNet merged-jet $\mathrm{H} \to \mathrm{c\bar{c}}$-tagging




Events vs. $\log_{10}(S/B)$

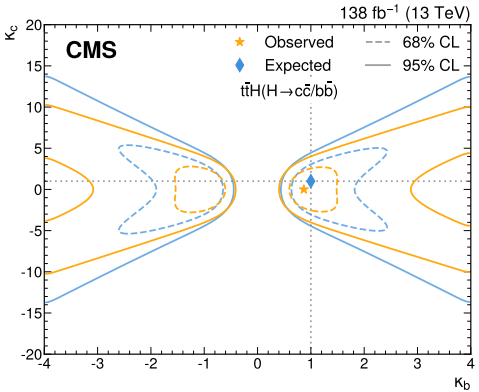



$|\kappa_c|$ scan (supplementary)

Best fit $\mu = 7.7^{+3.8}_{-3.5}$; merged $\mu = 8.7^{+4.6}_{-4.0}$, resolved $\mu = -9.5 \pm 9.6$. $\sigma\mathcal{B} < 0.94$ (0.50) pb.

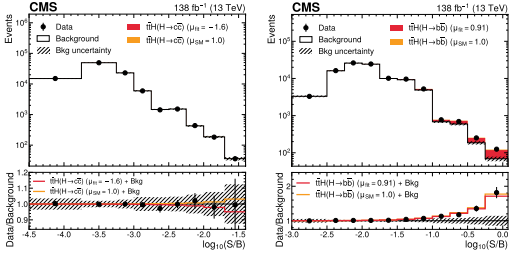
# $\mathrm{t\bar{t}H(H \to c\bar{c}/H \to b\bar{b})}$: $(\kappa_b, \kappa_c)$ plane

PRL 136 (2026) 011801

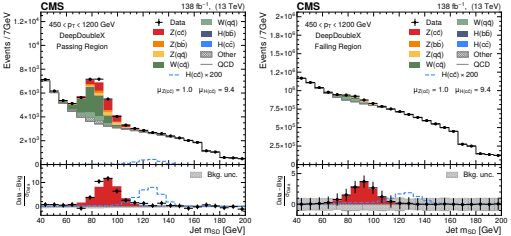



- Simultaneous $\mathrm{t\bar{t}H(H \to b\bar{b})}$ + $\mathrm{t\bar{t}H(H \to c\bar{c})}$ fit constrains both Yukawas
- Signal strengths: $\mathrm{t\bar{t}H}$(bb) $0.91^{+0.26}_{-0.22}$; $\mathrm{t\bar{t}H}$(cc) $-1.6 \pm 4.5$; $\mathrm{t\bar{t}Z}$(bb) $1.47^{+0.45}_{-0.41}$; $\mathrm{t\bar{t}Z}$(cc) $1.02^{+0.79}_{-0.84}$

# $\mathrm{t\bar{t}H(H \to c\bar{c})}$ and boosted $\mathrm{H \to c\bar{c}}$: more figures

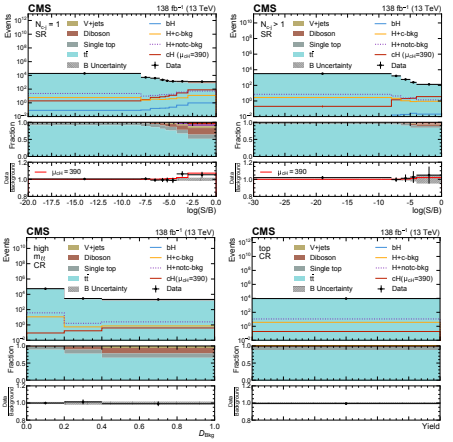



PRL 136 (2026) 011801: event yields vs. $\log_{10}(S/B)$

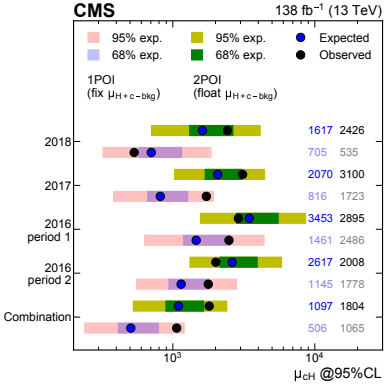



PRL 131 (2023) 041801: boosted $\mathrm{H \to c\bar{c}}$-jet mass, passing region

# cH(H→WW): post-fit distributions and per-period limits




1-POI post-fit distributions of the BDT discriminants

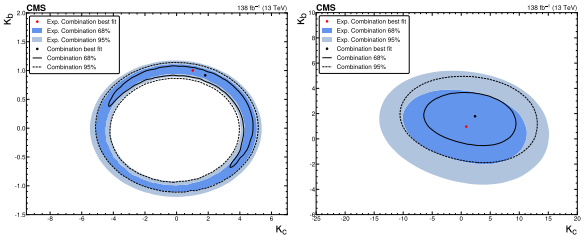



$\mu_{\mathrm{cH}}$ limits per data-taking period, 1 and 2 POI

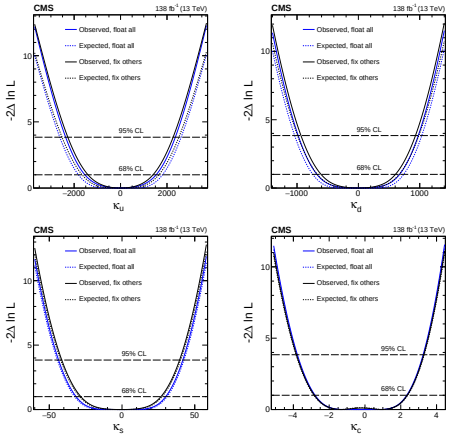
JHEP 08 (2026) 051

Best fit (1 POI): $\mu_{\mathrm{cH}} = 390^{+380}_{-270}$, compatible with SM within $1.5\sigma$; limited by data statistics, then flavour-scheme and ggH heavy-flavour modelling.

# Indirect: Higgs $p_{\mathrm{T}}$ spectrum and $\gamma$H




JHEP 04 (2026) 093: ($\kappa_{\mathrm{b}}$, $\kappa_{\mathrm{c}}$) from the Run 2 $\rho_{\mathrm{T}}^{\mathrm{H}}$ combination ($\gamma\gamma$, ZZ, WW, $\tau\tau$); left $\mathcal{B}$ coupling-dependent, right $\mathcal{B}$ unconstrained. Shape sensitive to the sign of $\kappa_{\mathrm{c}}$.

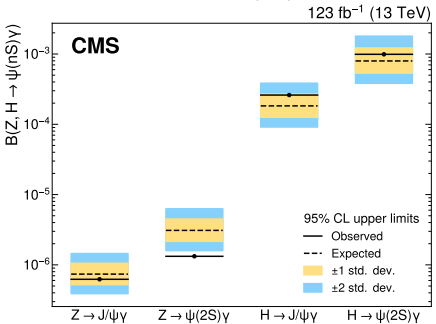



PRD 112 (2025) 112001: light-quark $\kappa$ scans

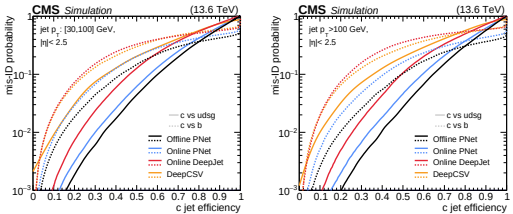
# Rare exclusive decays: $H \to J/\psi\gamma$, $\psi(2S)\gamma$

- Direct ($\propto \kappa_c$) and indirect ($H \to \gamma\gamma^* \to \gamma\, J/\psi$) amplitudes interfere $\Rightarrow$ sign sensitivity
- $123\,\mathrm{fb}^{-1}$: $\mathcal{B}(H \to J/\psi\gamma) < 2.6 \times 10^{-4}$, $\mathcal{B}(H \to \psi(2S)\gamma) < 9.9 \times 10^{-4}$
- $\kappa_c/\kappa_\gamma \in (-157, 199)$; with $\kappa_\gamma = 1$: $\kappa_c \in (-166, 208)$

PLB 865 (2025) 139462, CMS-SMP-22-012

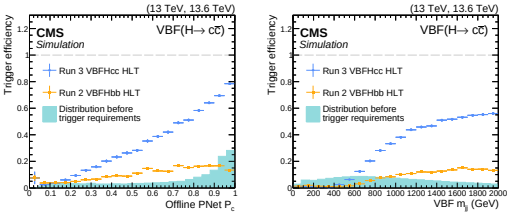


# Run 3 HLT flavour tagging




HLT tagger misidentification vs. efficiency

arXiv:2608.18331, CMS-BTV-25-002




VBF H→cc̄ trigger efficiency

# References


- VH($\mathrm{H} \to \mathrm{c\bar{c}}$): CMS, PRL 131 (2023) 061801, arXiv:2205.05550 (HIG-21-008)
- $\mathrm{t\bar{t}H}(\mathrm{H} \to \mathrm{c\bar{c}})$ + VH combination: CMS, PRL 136 (2026) 011801, arXiv:2509.22535 (HIG-24-018)
- Boosted $\mathrm{H} \to \mathrm{c\bar{c}}$: CMS, PRL 131 (2023) 041801, arXiv:2211.14181 (HIG-21-012)
- cH(H$\to \gamma\gamma$): CMS, JHEP 11 (2025) 060, arXiv:2503.08797 (HIG-23-010)
- cH(H$\to$WW) and combination: CMS, JHEP 08 (2026) 051, arXiv:2508.14988 (HIG-24-009)
- Differential combination Run 2 (backup): CMS, JHEP 04 (2026) 093, arXiv:2504.13081 (HIG-23-013)
- $\gamma$H + H$\to 4\ell$ light Yukawas: CMS, PRD 112 (2025) 112001, arXiv:2502.05665 (HIG-23-011)
- Off-shell + on-shell H$\to 4\ell$: CMS, arXiv:2607.23352, submitted to PRD (HIG-25-003)
- H$\to$J/$\psi\gamma$: CMS, PLB 865 (2025) 139462, arXiv:2411.15000 (SMP-22-012)
- HLT flavour tagging: CMS, arXiv:2608.18331 (BTV-25-002)
- $\mathrm{t\bar{t}}$+b/c cross sections: CMS-PAS-TOP-26-003
- HL-LHC: ATLAS and CMS, arXiv:2504.00672 (HIG-25-002)
- Higgs combination (motivation figure): CMS, Rep. Prog. Phys. 89 (2026) 087801, arXiv:2602.18611 (HIG-21-018)
- BSM: arXiv:2410.05236